Python Week 3 综合练习题

list

In [63]:
```
# 创建一个混合类型列表 info，包含："Alice"、22、1.65、"student".
info=["Alice",22,1.65,"student"]

# 访问列表 info 中第 3 个元素（索引 2）并打印
print(info[2])
```

```
1.65
```

In [64]:
```
# 修改 info 中第 2 个元素（索引 1）为 23
info[1]=23
print(info)
```

```
['Alice', 23, 1.65, 'student']
```

In [65]:
```
# 用 append() 在 info 末尾添加 "Malaysia"
info.append("Malaysia")
```

In [66]:
```
print(info)
```

```
['Alice', 23, 1.65, 'student', 'Malaysia']
```

In [67]:
```
# 用 insert() 在索引 0 位置插入 "Name:"
info.insert(0,"Nmae:")
```

In [68]:
```
print(info)
```

```
['Nmae:', 'Alice', 23, 1.65, 'student', 'Malaysia']
```

In [69]:
```
# 用切片取出 info 中索引 1 到 4 的元素（左闭右开）
info[1:5]
```

Out[69]:
```
['Alice', 23, 1.65, 'student']
```

In [70]:
```
# 删除 info 中值为 "student" 的元素
del info[4]
print(info)
```

```
['Nmae:', 'Alice', 23, 1.65, 'Malaysia']
```

In [71]:
```
# 用 pop() 删除 info 最后一个元素并打印被删除的值
print(info.pop())
```

```
Malaysia
```

二、for 循环与 range ()

In [72]:
```
# 遍历列表 fruits = ["apple","banana","orange"]，逐个打印水果名
fruits=["apple","banana","orange"]
```

```
for fruit in fruits:
    print(fruit)
```

```
apple
banana
orange
```

In [85]:
```
# 用 range() 生成 1 到 10 的数字，并存入列表 nums
nums=list(range(1,11))
print(nums)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

In [99]:
```
# 遍历 1 到 10，计算每个数的平方，存入列表 squares 并打印
squares=[]
for num in nums:
    squares.append(num**2)
print(squares)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

In [103…
```
# 遍历 nums 的前 3 个元素（切片）并打印
for n in nums[:3]:
    print(n)
```

```
1
2
3
```

三、if 条件判断

In [ ]:
```
# 输入学生分数（0-100）
# 90 及以上 → 输出 A
# 80-89 → B
# 70-79 → C
# 60-69 → D
# 60 以下 → F
```

In [12]:
```
mark=float(input("please input student's marks:"))
if mark >= 90:
    print("A")
elif 80<= mark <=89:
    print("B")
elif 70<= mark <=79:
    print("C")
elif 60<= mark <=69:
    print("D")
else:
    print("F")
```

```
C
```

In [ ]:
```
# 如果分数 >100 或 <0，输出 Invalid score
# 否则按等级输出：
# ≥85 → A
# ≥75 → B
# ≥60 → C
# <60 → F
```

In [27]:
```
mark=float(input("please input student's marks:"))
if mark<0 or mark>100:
    print("Invalid score")
elif mark<60:
    print("F")
elif 60<=mark<75:
    print("C")
elif 75<=mark<85:
    print("B")
else:
    print("A")
```

```
Invalid score
```

In [23]:
```
mark=float(input("please input student's marks:"))
if mark<0 or mark>100:
    print("Invalid score")
elif mark>=85:
    print("A")
elif mark>=75:
    print("B")
elif mark>=60:
    print("C")
else:
    print("F")
```

```
C
```

In [104…
```
# 判断变量 score = 85 是否大于等于 60，打印"及格"或"不及格"
score=85
if score >=60:
    print("PASS")
else:
    print("FALI")
```

```
PASS
```

In [111…
```
# 判断 num = 18 是否在列表 check = [12,15,18,21] 中
check=[12,15,18,21]
print(18 in check)
```

```
True
```

In [113…
```
age=19
if age<5:
    print("FREE")
elif age<18:
    print("5马币")
else:
    print("10马币")
```

```
10马币
```

In [122…
```
# 判断 a=10 是否大于 5 且小于 20
a=10
print(5<a<20)
```

```
True
```

In [121…
```
# 判断 "cat" 是否不在列表 pets = ["dog","rabbit"] 中
pets=["dog","rabbit"]
```

```
print("cat" not in pets)
```

```
True
```

四、字典（Dictionary）

In [123…

```
# 创建字典 student，包含键值对："name":"Tom"、"age":20、"grade":"A"
student={"name":"Tom","age":20,"grade":"A"}
print(student)
```

```
{'name': 'Tom', 'age': 20, 'grade': 'A'}
```

In [124…

```
# 打印 student 中 "name" 对应的值
print(student["name"])
```

```
Tom
```

In [125…

```
# 添加键值对 "city":"KL" 到 student
student["city"]="KL"
print(student)
```

```
{'name': 'Tom', 'age': 20, 'grade': 'A', 'city': 'KL'}
```

In [126…

```
# 修改 student 中 "age" 为 21
student["age"]=21
print(student)
```

```
{'name': 'Tom', 'age': 21, 'grade': 'A', 'city': 'KL'}
```

In [127…

```
# 删除 student 中 "grade" 键值对
del student["grade"]
print(student)
```

```
{'name': 'Tom', 'age': 21, 'city': 'KL'}
```

In [132…

```
# 遍历 student 所有键 + 值并打印
for key,value in student.items():
    print("\nKey:"+key)
    print("Value:"+str(value))
```

```
Key:name
Value:Tom

Key:age
Value:21

Key:city
Value:KL
```

In [133…

```
for k,v in student.items():
    print(k,v)
```

```
name Tom
age 21
city KL
```

五、用户输入 input ()

In [142…

```
name=input("name:")
age=int(input("age:"))
print("Hello "+name+", you are "+str(age)+" years old!")
```

```
name: Mica
age: 20
Hello Mica, you are 20 years old!
```

六、while 循环

In [143…

```
# 用 while 循环打印 1 到 5
n=1
while n<6:
    print(n)
    n+=1
```

```
1
2
3
4
5
```

In [5]:

```
# 循环到 3 时用 break 停止
n=1
while n<6:
    if n==3:
        break
    print(n)
    n+=1
```

```
1
2
```

In [3]:

```
# 循环跳过 3，打印 1、2、4、5
n=0
while n<5:
    n+=1
    if n==3:
        continue
    print(n)
```

```
1
2
4
5
```

In [7]:

```
# 用 flag 标志位控制循环，从 10 倒数到 1 停止
n=10
flag=True
while flag:
    print(n)
    if n==1:
        flag=False
    n-=1
```

```
10
9
8
7
6
5
4
3
2
1
```

七、综合嵌套

In [13]:
```
people=[
    {"name":"Lily","age":19},
    {"name":"Lucas","age":22}
]
for p in people:
    print(p)
```

```
{'name': 'Lily', 'age': 19}
{'name': 'Lucas', 'age': 22}
```

In [15]:
```
car={
    "Proton":["Saga","X50"],
    "Perodua":["Myvi","Axia"]
}
for k,v in car.items():
    print(k,v)
```

```
Proton ['Saga', 'X50']
Perodua ['Myvi', 'Axia']
```

专项强化练习题

In [18]:
```
city=["Beijing","Shanghai"]
city.insert(0,"Country:")
print(city)
```

```
['Country:', 'Beijing', 'Shanghai']
```

In [20]:
```
numbers=list(range(2,12))
print(numbers)
```

```
[2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
```

In [26]:
```
numbers=list(range(2,9))
sq=[]
for num in numbers:
    sq.append(num**2)
print(sq)
```

```
[4, 9, 16, 25, 36, 49, 64]
```

In [30]:
```
nums=[10,20,30,40,50]
for n in nums[:3]:
    print(n)
```

```
10
20
30
```

In [32]:
```
x=22
lst=[10,15,20,25,30]
print(x in lst)
```

```
False
```

In [7]:
```
n=0
while n<6:
    n += 1
    if n==4:
        continue
    print(n)
```

```
1
2
3
5
6
```

In [1]:
```
n=8
flag=True
while flag:
    print(n)
    if n==1:
        flag=False
    n -= 1
```

```
8
7
6
5
4
3
2
1
```

while + continue

In [3]:
```
i=0
while i<6:
    i += 1
    if i == 4:
        continue
    print(i)
```

```
1
2
3
5
6
```

In [4]:
```
i = 0
while i < 7:
    i += 1
    if i==6:
```

```
        continue
    print(i)
```

```
1
2
3
4
5
7
```

In [6]:
```
i=1
while i<10:
    i+=1
    if i==8:
        continue
    print(i)
```

```
2
3
4
5
6
7
9
10
```

while + flag

In [7]:
```
flag=True
i=7
while flag:
    print(i)
    if i==1:
        flag=False
    i -= 1
```

```
7
6
5
4
3
2
1
```

In [8]:
```
flag=True
n=12
while flag:
    print(n)
    if n==1:
        flag=False
    n -= 1
```

```
12
11
10
9
8
7
6
5
4
3
2
1
```

In [9]:
```python
flag=True
x=5
while flag:
    print(x)
    if x==1:
        flag=False
    x -=1
```

```
5
4
3
2
1
```

In [ ]:
```python
# 用 while 循环 + flag + continue
#从 5 倒数到 1
#跳过数字 3
#输出: 5 4 2 1
```

In [1]:
```python
flag = True
i = 5

while flag:
    if i == 3:
        i -= 1
        continue
    print(i)
    if i == 1:
        flag = False
    i -= 1
```

```
5
4
2
1
```

In [2]:
```python
flag=True
i=8

while flag:
    if i==5:
        i -= 1
        continue
    print(i)
    if i==1:
        flag=False
    i -=1
```

```
8
7
6
4
3
2
1
```

In [ ]:
```
#用 while 循环 + flag + continue
#从 10 倒数到 1
# 跳过数字 7 和 4
```

In [3]:
```
flag=True
i=10

while flag:
    if i==7 or i==4:
        i-=1
        continue
    print(i)

    if i==1:
        flag=False
    i -=1
```

```
10
9
8
6
5
3
2
1
```

In [ ]:
```
#用 while + flag 循环
#从 1 数到 10
#跳过 5
#数到 10 时结束循环
#输出: 1 2 3 4 6 7 8 9 10
```

In [4]:
```
flag=True
i=1

while flag:
    if i==5:
        i+=1
        continue
    print(i)
    if i==10:
        flag=False
    i+=1
```

```
1
2
3
4
6
7
8
9
10
```

In [ ]:
```
#用 while + flag 循环
#从 1 数到 12
#同时跳过 3 和 9
#数到 12 结束
#输出: 1 2 4 5 6 7 8 10 11 12
```

In [6]:
```
flag=True
i=1

while flag:
    if i==3 or i==9:
        i+=1
        continue
    print(i)

    if i==12:
        flag=False
    i+=1
```

```
1
2
4
5
6
7
8
10
11
12
```

In [1]:
```
flag=True
i=1

while flag:
    if i == 5 or i == 10 or i == 15:
        i += 1
        continue

    print(i)

    if i == 14:  # 输出完14就结束
        flag=False
    i += 1
```

```
1
2
3
4
6
7
8
9
11
12
13
14
```

In [ ]:
```
1+1
```

In [5]:
```
# 使用 while + flag，打印 1~7，跳过 4
flag=True
i=1
while flag:
    if i==4:
        i+=1
        continue
    print(i)

    if i==7:
        flag=False
    i+=1
```

```
1
2
3
5
6
7
```

In [1]:
```
# 使用 while + flag，打印 1~9，跳过 7
flag=True
i=1
while flag:
    if i==7:
        i+=1
        continue
    print(i)

    if i==9:
        flag=False
    i+=1
```

```
1
2
3
4
5
6
8
9
```

In [2]:
```
# 使用 while + flag，打印 1~12，跳过 2 和 9
flag=True
i=1
```

```
while flag:
    if i==2 or i==9:
        i+=1
        continue
    print(i)

    if i==12:
        flag=False
    i+=1
```

```
1
3
4
5
6
7
8
10
11
12
```

In [3]:

```
# while + flag，从 10 倒数到 1，跳过 6
flag=True
i=10

while flag:
    if i==6:
        i-=1
        continue
    print(i)

    if i==1:
        flag=False
    i-=1
```

```
10
9
8
7
5
4
3
2
1
```

In [4]:

```
# while + flag，从 1 数到 15，跳过 5、10、15
flag=True
i=1

while flag:
    if i==5 or i==10 or i==15:
        i+=1
        continue
    print(i)

    if i==14:
        flag=False
    i+=1
```

```
1
2
3
4
6
7
8
9
11
12
13
14
```

In [6]:
```python
# while + flag，从 1 数到 20，跳过所有 3 的倍数（3、6、9…）
flag=True
i=1

while flag:
    if i%3==0:
        i+=1
        continue
    print(i)

    if i==20:
        flag=False
    i+=1
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
```

In [ ]:
```python
# while + flag，1~10:
# 偶数打印
# 奇数跳过
```

In [8]:
```python
flag=True
i=1

while flag:
    if i%2 !=0:
        i+=1
        continue
    print(i)

    if i==10:
        flag=False
    i+=1
```

```
2
4
6
8
10
```

In [ ]:
```
# while + flag, 1~10:
# 偶数跳过
# 奇数打印
```

In [2]:
```
flag=True
i=1

while flag:
    if i%2 ==0:
        i+=1
        continue
    print(i)

    if i==9:
        flag=False
    i+=1
```

```
1
3
5
7
9
```

In [ ]:
```
# while + flag, 1~14:
# 跳过 5 和 10
# 打印到 14 自动结束
```

In [3]:
```
flag=True
i=1

while flag:
    if i==5 or i==10:
        i+=1
        continue
    print(i)

    if i==14:
        flag=False
    i+=1
```

```
1
2
3
4
6
7
8
9
11
12
13
14
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: